Ghost Data Audit

Project Report — HackerRank Orchestrate, “Ghost Data Still Exists”

Fact	Value
Entrant	Chirantan (solo)
Repository	github.com/Cshalya30/Hackerrank-Orchestrate
Report generated	2026-10-25
Final sample scores	Status accuracy 0.812 · Step F1 0.706 · Sequence accuracy 0.156
Full run	518/518 rows valid · COMPLETE 211 · INCOMPLETE 297 · NOT_APPLICABLE 10
LLM usage	17 calls · 44,080 tokens · ~$0.006 at list price
Commits	65 total on main

This report covers the full build: the evidence-first deletion-audit pipeline, every kept and reverted design decision, the evaluation history end to end, an LLM-usage accounting, two incidents found and resolved mid-session (a lost-logging recovery and a live parallel-session race), and the three-iteration design journey of the pipeline visualizer. Nothing here is invented — every number traces to a file in the repository.

1. Executive Summary

The challenge asks for an audit agent that reads 518 data-deletion requests against a 94-file multi-system evidence corpus and, for each request, decides COMPLETE / INCOMPLETE / NOT_APPLICABLE and emits an ordered FINDING → REMEDIATION → VERIFICATION step list with evidence citations. The approach taken is evidence-first and almost entirely deterministic: the shared evidence corpus is parsed once into a flat observation log, each request's identity is resolved to its full historical footprint, resource state is reconstructed at the point in time of the audit, and the 15 deletion rules are applied as plain Python — not as LLM reasoning. The only non-deterministic step is request-intent classification, which genuinely needs natural-language judgment across six-plus languages.

Final, reproducible sample-set scores are **0.812** status accuracy, **0.706** step F1 (precision 0.807 / recall 0.627), and **0.156** sequence accuracy. The full 518-row run validates cleanly against the submission contract. Two screenshot(vision-LLM) merge attempts were tried and ultimately reverted because their effect on the score was not reproducible across runs — reproducibility was weighted over an unconfirmed score gain, in line with the project's own determinism requirement.

Two real incidents occurred mid-build and are documented in full below: a session that ran from the wrong working directory and lost live logging (recovered from the harness's own session transcript), and a second session that never actually terminated and kept editing the same repository in parallel — caught via file-change detection and a process-list check, then reconciled by verifying every one of its claims against the current code rather than trusting its commit message.

Key numbers at a glance

Metric	First build	Final (reproducible)
Status accuracy	0.531 → 0.750 (NOT_APPLICABLE bug fixed)	0.812
Step F1	0.529	0.706
Step precision / recall	0.880 / 0.378	0.807 / 0.627
Sequence accuracy	0.062	0.156
Coverage (evidence pairs)	82.0%	91.0%
Full-run NOT_APPLICABLE count	6 (keyword fallback)	10 (LLM, matches advisor estimate)

2. Problem Statement

518 data-deletion requests, each tied to a user by email. For every request the system must decide a **deletion_status** (COMPLETE, INCOMPLETE, or NOT_APPLICABLE) and emit an ordered JSON list of steps over (system_id, resource_id) pairs, each step one of FINDING / REMEDIATION / VERIFICATION, citing the evidence file that justifies it.

The evidence corpus

Source	Count	Notes
Evidence files total	94	24 SQL dumps, 28 JSON + 1 TXT, 26 logs, 4 PDF, 11 PNG screenshots
user_identities.csv	850	current + historical identifiers per user
user_audit_log.csv	306	identity change history
system_activity.csv	4,011	activity claims (deletes, links, restores, etc.)
deletion_rules.csv	15	the policy table — category → required treatment
Public samples	32	0 NOT_APPLICABLE, 22 INCOMPLETE, 10 COMPLETE, gold-labeled

The core mechanic: “ghost data”

A resource can be proven ABSENT by file evidence at one point in time, then resurrected by a later successful activity (e.g. a backup restore) — the system must resolve to the state as of each request's specific **audit_at** timestamp, not simply “latest known state.” This point-in-time reconstruction, combined with the rule that file evidence outranks activity claims (claims are not proof), is the central mechanic the evaluation rewards.

3. System Architecture

Nine stages, almost entirely deterministic. The pipeline visualizer (`tools/viz/index.html`) renders this exact graph live from build artifacts — nothing in the diagram below is hand-maintained separately from the code.

#	Stage	File :: function	Kind
1	Deletion Requests	dataset/deletion_requests.csv	Data
2	Intent (LLM)	code/intent.py :: classify_case	LLM
3	Identity Resolution	code/resolve.py :: IdentityIndex.resolve_user	Deterministic
4	Evidence Ingest	code/ingest/build.py :: main	Deterministic
5	Point-in-time State	code/resolve.py :: latest_state_per_resource	Deterministic
6	Policy / Account Rules	code/account_linker.py :: AccountLinker.account_for	Deterministic
7	Planner	code/planner.py :: build_steps / finalize	Deterministic
8	Validator	code/validate.py :: main	Validation
9	output.csv	—	Data

Why the LLM is scoped to one stage

Gemini Flash-Lite (`gemini-flash-lite-latest`, thinking off) is used for exactly one thing: request-intent classification, batched 25 distinct `request_text` strings per call, disk-cached by prompt hash. This is the one genuinely open-ended, multilingual task in the system — request text appears in at least six languages and includes subtle “verify my deletion” follow-ups that a keyword list cannot generalize to. Every other stage is deterministic Python by design: a judge re-running the exact reproduce commands should get the exact same `output.csv` every time.

Intent labels

- **DELETE_REQUEST** — a direct ask to delete/close the account or erase personal data now.
- **DELETION_VERIFICATION** — the person already asked for deletion and is now asking to verify/confirm/audit it — still pursuing removal of their own data, so it stays in the deletion pipeline, not NOT_APPLICABLE.
- **OTHER_SERVICE** — anything else (update, export, reset, invoice, email change, “keep my account”) — the only label that maps to NOT_APPLICABLE.

A `CLASSIFY_REQUEST` activity override always forces OTHER_SERVICE regardless of the LLM's own read of the text, and a hard gate on the sample split fails the run if it ever produces a NOT_APPLICABLE case (the gold sample set has zero, by design).

4. Key Design Decisions

Every decision below is logged with its alternative and its evidence in `docs/DECISIONS.md`; reverted attempts are kept in the record, not deleted, so the reasoning trail survives.

#	Decision
D1	**Evaluation artifacts live under code/evaluation/, not root evaluation/** README.md's packaging section and the starter repo's existing code/evaluation/main.py stub both confirm this placement over the original SPEC draft.
D2	**Runtime LLM is scoped to intent/image/rare-adjudication only** Everything else (identity, state, policy, step order) is deterministic Python. LLM-per-case reasoning was rejected as expensive, non-deterministic, and harder to defend in the AI interview.
D3	**Point-in-time state: file evidence outranks activity claims; ties favor PRESENT** Per (system_id, resource_id), the latest observation at or before audit_at wins; on an exact tie, PRESENT beats ABSENT, modeling replica drift rather than assuming the best case.
D4	**One consolidated code/ingest/build.py instead of five parser modules** The originally sketched sql_parser.py / json_parser.py / log_parser.py / text_parser.py split would have shared nearly all helper logic; consolidated under time budget.
D5	**system_activity.csv is a required ingest source, not just an overlay** First ingest pass hit only 82.0% coverage, every miss an account_service row; account_id numeric suffixes do not match user_id suffixes, so these accounts are only discoverable via system_activity.csv's LINK_RESOURCE_IDENTITY rows or DELETE_RECORD claims. Coverage rose to 90.1% once wired in.
D6	**Empty resource_states on a DELETION case means COMPLETE with [], never NOT_APPLICABLE** The first implementation conflated “no steps found” with “not a deletion request,” producing 11 confidently-wrong NOT_APPLICABLE predictions against a gold set with zero. Fixing this raised status accuracy 0.531 → 0.750.
D7	**Reverted: excluding decoy prefixes (object_storage) from LINK_RESOURCE_IDENTITY** Fixed one confirmed false positive (case_07) but flipped 0 cases net and relied on a coincidence elsewhere; F1 unchanged, status accuracy regressed 0.812 → 0.781. Reverted per the “keep only if F1 improves” rule.
D8	**Reverted: merging all 11 image_parser.py observations** All 11 screenshots parsed cleanly into 34 observations, but merging them regressed sample F1 0.706 → 0.695 — some vision-extracted rows appear to shadow correct file-evidence states under the PRESENT-beats-ABSENT tie rule.
D9	**Kept (provisionally): a 9/11 partial image merge net-improved the score** A later re-run only parsed 9/11 screenshots (2 hit a free-tier daily quota, not a parsing bug); this smaller merge raised status accuracy to 0.844 and F1 to 0.720 — the opposite of D8. Flagged as a hypothesis, not confirmed, pending a reproducibility check.
D10	**Re-reverted: D9's result does not reproduce — vision-LLM non-determinism** Re-running with the daily quota reset (all 11/11 succeeded) reproduced D8's regression exactly, not D9's gain. Hardcoding the same 2-file exclusion that D9 credited did not reproduce 0.844/0.720 either — the same cached image returned a different observation count across separate calls. Removed the merge entirely; reproducibility was weighted over an unconfirmed +0.014 F1 gain, per the project's own determinism requirement.

5. Evaluation History

Every sample-set eval run this session, kept and reverted, in order. Format: status accuracy / step F1 / sequence accuracy against the 32 public samples.

Time	Change	Status	F1	Seq.
20:52	sample_output.csv vs itself (eval.py sanity check)	1.000	1.000	1.000
21:40	First Card 4 run: identity + state + policy + planner, no fixes	0.531	0.529	0.062
21:48	Fix: finalize() no longer infers NOT_APPLICABLE from empty steps	0.750	0.529	0.062
00:15	RUN2 Group A: account_linker.py, audit record wiring	0.750	0.497	0.156
00:30	Fix: 404 status_code check, field extraction, resource-continuity overlay	0.750	0.646	0.156
01:00	Fix: category-aware default states, email_hash matching, MARKETING_PROFILE	0.812	0.694	0.156
01:15	B5: PDF parser wired for support attachment + audit record	0.812	0.706	0.156
23:50	3-label intent scheme on Gemini Flash-Lite; sample gate added	0.812	0.706	0.156
00:05	TRIED → REVERTED: 11/11 image merge regressed F1	0.812	0.706	0.156
09:40	KEPT (provisional): 9/11 image merge net-improved	0.844	0.720	0.156
09:45	Intent switched to Gemini LLM path confirmed, no regression	0.844	0.720	0.156
10:20	Phase-2 re-check: 11/11 re-run reproduced the D8 regression, not D9	0.812	0.695	0.156
10:25	Hardcoded 9/11 exclusion did NOT reproduce D9 — re-reverted	0.812	0.695	0.156
10:30	Final, reproducible state: build.py alone, no image merge	0.812	0.706	0.156

Field-level accuracy (final state)

Field	Accuracy
step_no	1.000
phase	0.815
system_id	0.582
resource_id	0.541
operation	0.562
state	0.616
evidence_id	0.884

6. Known Failures and the Fixes Made

#	Failure → Fix	Effect
1	**eval.py's field-accuracy divisor summed all 7 fields into one shared total** Fix: Per-field total dict instead of one shared counter	All 7 field accuracies correctly read 1.000 on the self-check
2	**planner.finalize() collapsed "zero resources resolved" into NOT_APPLICABLE** Fix: Only intent classification may set NOT_APPLICABLE; empty-resource cases become COMPLETE with []	Status accuracy 0.531 → 0.750
3	**First ingest pass covered only 82.0% of required evidence pairs** Fix: Parse system_activity.csv (LINK_RESOURCE_IDENTITY, DELETE_RECORD, RESTORE_BACKUP)	Coverage 82.0% → 90.1%
4	**Deterministic intent fallback defaulted 6 unambiguous non-deletion requests to DELETION** Fix: Broadened the non-deletion keyword hint list	Full run: 0 → 6 correct NOT_APPLICABLE
5	**Many account/audit resources never resolved: 404 status_code never checked, nested fields never extracted, a closure late-binding bug corrupted every export's source field** Fix: account_linker.py priority chain + four parser bug fixes	Step F1 0.529 → 0.646
6	**Log/activity sources defaulted to generic PRESENT; old email-hash-only matches were missed** Fix: Category-aware default states; identity matching extended to email_hash/email sets	Status 0.750 → 0.812, F1 0.646 → 0.694
7	**support_001/audit_record evidence only existed in a PDF, unparsed** Fix: Wired a targeted pdf_parser for the two specific PDFs needed	Step F1 0.694 → 0.706
8	**Intent classification was binary and keyword-based; verification follow-ups were indistinguishable from ambiguous defaults** Fix: 3-label scheme on Gemini Flash-Lite, batched/cached	Full-run NOT_APPLICABLE count 6 → 10, matching the advisor's 10–15 estimate
9	**Screenshot evidence parsing tried twice, reverted both times** Fix: Not merged — vision-LLM extraction is not reproducible call to call	Final numbers (0.812/0.706) are the real, reproducible state

Remaining known gaps

- **Screenshot evidence is not merged** into the pipeline — the vision model's per-call row extraction is not fully deterministic, so its net effect on F1 cannot be trusted across repeated runs (see D8/D9/D10). A future session should diff extraction output across several repeated calls on the same image before re-attempting a merge.
- **7 accounts and 3 audit records have zero evidence anywhere** in the parsed corpus, even after account_linker's batch-matching and zero-padded fallback.
- **SUPPORT_ATTACHMENT vs. SUPPORT_MESSAGE** is not fully reliable beyond two PDF-backed special cases — the general heuristic still defaults by table/collection name.
- **The deterministic intent fallback** (used only when no API key is configured) will not generalize to unseen non-deletion or verification phrasing the way the LLM path does.
- **157 full-run cases** have no resolvable account_service id even after every fallback in the resolution chain.

7. LLM Usage and Cost

Final, clean-sequence usage for the committed pipeline (sample-split gate + full run).

Model	Calls	Cached	Input tok.	Output tok.	Total tok.	Est. cost
gemini-flash-lite-latest	17	0	32,236	11,844	44,080	$0.005971

Average per request (of 518 full-dataset requests): 85.1 tokens, $0.0000115.

What each call is for

- **Intent classification** — one batch call per 25 distinct request_text strings (2 batches for the 32 sample cases + 15 for 375 distinct full-dataset texts = 17 calls). Cached per batch by prompt hash.
- **Screenshot parsing** (not in the committed pipeline) — one call per PNG, tried twice as an experiment and excluded from the table above since it was not kept.

8. Incidents Found and Resolved Mid-Session

8.1 Lost logging: session ran from the wrong working directory

A prior continuation session ran with its working directory set to an unrelated project folder instead of this repository. Its Run 2 work (account/audit resolution, category-aware states, PDF wiring) produced real commits but no live log.txt entries, since the AGENTS.md-mandated log file could not be found/updated from the wrong location.

Resolution: built `tools/recover_log.py`, which reads the coding harness's own session transcript (.jsonl) directly, extracts each missed user turn and its tool actions, and reconstructs log.txt entries with their original timestamps, explicitly marked `recovered=true`. A header entry documents the recovery itself. Verified by cross-checking the recovered entries' described work against the actual git commits from that window.

8.2 Parallel-session race: an old session never actually stopped

While a new, correctly-located session was working, files began changing on disk (docs/EVAL_LOG.md, evaluation_report.md, usage_report.md, requirements.txt, a test assertion) without any edit from the active session. Investigation found two `image_parser.py` processes still running in the background, started by the earlier misplaced-directory session, which had never terminated.

Resolution: work was paused immediately and the situation was reported to the owner rather than racing the other session or guessing at a merge. The owner confirmed the old session and closed its terminal. Rather than trusting the other session's commit message, every one of its claims was independently re-verified against the current code and a fresh eval re-run before anything was kept — which is how the D9/D10 non-reproducibility finding above was ultimately surfaced: the “kept” 0.844/0.720 result did not survive a second, independent verification pass.

Lesson applied: the project's own log.txt protocol and a simple process-list check (“is anything still pointing into this folder?”) were enough to catch a live concurrency hazard before it caused data loss — and treating a prior session's own commit message as a claim to verify, not a fact to trust, caught a genuine non-determinism bug that would otherwise have shipped as a false “improvement.”

9. The Pipeline Visualizer: a Three-Iteration Design Journey

Built for the live interview demo: `tools/viz/export_stats.py` reads build artifacts (observations, traces, the LLM ledger, output.csv, the sample eval) into `pipeline.json`, and `tools/viz/index.html` renders it as a single self-contained page — no build step, no framework, no external requests beyond one Google Fonts stylesheet.

v1 — Glow/glass canvas — Rejected

A dark canvas with ambient drifting gradient blobs, glassmorphic node cards, emoji kind-icons, and particle-flow edges. Functionally complete (pan/zoom, case trace, node detail panel) but called out as generic “AI-SaaS” styling — the exact clichés (purple gradient hero, glassmorphism, emoji icons) a design-quality skill explicitly warns against.

v2 — Forensic “case docket” theme — Rejected

A full redesign grounded in the subject itself (a deletion audit is literally a forensic/compliance artifact): warm ink palette, a single ochre case-stamp accent, Fraunces serif + monospace typography, exhibit-stamped node cards, a numbered step ledger. Caught and fixed two real bugs in the process (a hardcoded header-height assumption that pushed content off-screen, and an over-granular by_source bucketing bug producing ~3,000 near-duplicate categories). Still rejected — typography, palette, and the forensic theme itself all read wrong to the owner, and it still felt AI-generated overall.

v3 — Stripe-register dashboard — Rejected (direction changed)

Asked three clarifying questions before guessing a third time (what specifically was wrong, which real product's register to match, whether to keep the node-canvas or restructure). Built a conventional light-mode operations dashboard — KPI cards, a pipeline stepper, a searchable/paginated case table with a slide-in drawer — matching the owner's named Stripe reference exactly. Fully verified (3 breakpoints, every interactive path exercised). The owner then clarified the node-flow canvas itself was the part they liked — the dashboard replaced the wrong thing.

v4 — Premium maximalist flow canvas (final) — Shipped

Reinstated the node-flow canvas and applied the motion-design skill's “Premium” motion personality (350–600ms, cubic-bezier(.4,0,.2,1), zero overshoot, three motion layers) deliberately rather than freehand: a fixed aurora-mesh ambient background, a bold Bricolage Grotesque stat marquee, gradient edges carrying continuously animated comet particles, and a “Trace case” button that lights a case's full path through the pipeline in sequence. Caught a real CSS bug via Playwright (not a screenshot alone): the trace animation's class swap silently stripped the node entrance animation's opacity hold, making traced nodes disappear — fixed by isolating the entrance into a one-shot class removed on animationend.

Bugs caught by actually testing the page, not just viewing a screenshot

Bug	How it was caught	Fix
Header wraps to 2 rows but layout used a hardcoded calc(100vh - 78px)	Playwright getBoundingClientRect() on the header vs. the assumed height	Switched body/header/main to a proper flex-column layout
by_source grouped ~3,000 near-unique per-file keys instead of ~10 categories	Garbled text visible in a ticker component and the node detail panel	Bucket on the category prefix before the first ':' in export_stats.py

Bug	How it was caught	Fix
Status legend truncated to "10 not_" (repeated twice across redesigns)	Visual inspection of the rendered header	Replaced string slicing with a real label lookup table
Traced nodes went invisible after the case-trace animation played	Checked computed opacity via Playwright immediately after firing the trace	Isolated the entrance animation into a removable .entering class

10. Custom Evaluation Cases

Three synthetic cases, each isolating one mechanism, run standalone with no pytest dependency (`code/evaluation/custom_cases/test_custom_cases.py`). All three pass.

Case	What it proves
case_1_not_applicable	A request mixing an export ask with explicit “leave my account as it is” framing must classify as OTHER_SERVICE → NOT_APPLICABLE.
case_2_old_identifier_match	A record carrying only the user's historical (pre-change) email hash must still be matched by identity resolution and planned for ANONYMIZE_RECORD.
case_3_resurrection_after_deletion	A resource proven ABSENT by file evidence at T1, then claimed PRESENT again by a later successful activity at T2, must resolve to PRESENT at any audit_at after T2 — the central “ghost data” mechanic.

11. Submission Package

Contents of submission/, built and secret-scanned by tools/make_zip.py:

File	Contents
code.zip	32 files: code/, tools/ (incl. the visualizer), docs/, root README.md, requirements.txt, .env.example. Excludes .venv, .env, build/, __pycache__, dataset/. Secret-scanned clean on every build.
output.csv	518 validated rows, the exact file code/validate.py last passed.
chat_transcript.txt	Full copy of log.txt — every session entry per AGENTS.md section 5, including the two recovered/reconciled incidents above.
SUBMISSION_CHECKLIST.md	Pre-submission verification list and known-gaps summary to raise proactively with the interviewer.
INTERVIEW_PREP.md	60-second pitch, architecture walkthrough, one answer each for 6 required topics, 5 failure stories, honest limitations, 10 hard Q&A;, a 90-second visualizer demo script.

Reproduce every number in this report

```
python code/ingest/build.py
python code/main.py --split sample --out build/sample_output_pred.csv
python code/evaluation/main.py --pred build/sample_output_pred.csv --gold
dataset/sample_output.csv
python code/main.py --split full --out output.csv
python code/validate.py output.csv --expect-rows 518 --case-ids dataset/deletion_requests.csv
python code/evaluation/custom_cases/test_custom_cases.py
```

Open the visualizer

```
python -m http.server 8765 -d tools/viz
# open http://localhost:8765
```

12. Development Timeline

65 commits on main, condensed to the major milestones:

When	What happened
Oct 23–24	Repo setup, dataset/problem-statement updates (organizer-side)
Oct 24, 20:25–22:40	Run 1: full pipeline skeleton, eval.py + validate.py, evidence profiler, ingest parsers (90.1% coverage), identity/state/planner, custom cases, evaluation + usage reports. Sample score: 0.750 / 0.529 / 0.062.
Oct 24, 23:10–23:18	Run 2 (uncommitted to log.txt live — recovered later): account_linker.py, category-aware states, PDF parsing. Score: 0.812 / 0.706 / 0.156.
Oct 24, 23:42–23:53	Continuation session: log recovery tool, wip Gemini intent + image parser committed as-is, 3-label intent scheme shipped, image-merge regression found and reverted (D8).
Oct 25, 00:07–00:20	Parallel-session incident discovered, reported, and reconciled; image-merge non-reproducibility found and the merge re-reverted (D10). Final numbers locked at 0.812 / 0.706 / 0.156.
Oct 25, 00:24–00:28	Pipeline visualizer v1 (glow/glass) built and shipped; code/README.md and final CONTEXT.md written; full submission package assembled and tagged v1-final.
Oct 25, 00:34–01:04	Visualizer redesigned three more times on owner feedback (case-docket → Stripe dashboard → premium maximalist canvas), each verified with Playwright before shipping; real bugs caught and fixed at every stage.

13. Conclusion

The final pipeline is evidence-first and deterministic everywhere except one necessarily fuzzy stage (intent classification), achieves 0.812 status accuracy and 0.706 step F1 on the public sample set, and validates cleanly across all 518 full-run cases. Every design decision — kept or reverted — is logged with its evidence, and a result that could not be reproduced on a second run (D9's 0.844/0.720) was caught and removed rather than shipped, even though it looked like a genuine improvement at first.

Two process incidents (lost logging from a misplaced working directory, and a live parallel-session race) were caught, reported honestly, and resolved without losing or fabricating any work. The interview-facing pipeline visualizer went through four design passes driven by direct feedback, each one actually tested rather than just rendered, surfacing four more real bugs along the way.

Known, honestly-stated gaps remain: screenshot evidence is parseable but not merged (non-deterministic), 157 full-run cases lack a resolvable account id, and SUPPORT_ATTACHMENT/SUPPORT_MESSAGE disambiguation is incomplete beyond two special cases. These are documented in docs/GAPS.md and code/evaluation/evaluation_report.md, not hidden, and are the first things raised proactively in the interview per submission/SUBMISSION_CHECKLIST.md.

Generated from the live contents of docs/DECISIONS.md, docs/EVAL_LOG.md, docs/GAPS.md, code/evaluation/evaluation_report.md, code/evaluation/usage_report.md, log.txt, and the project's git history. See code/README.md for full setup and run instructions.